

Analysing the Air Quality Index (AQI) in Delhi: Pollutants, Seasonal Variations, and Geographical Impacts

Introduction:

Delhi, the capital city of India, is frequently listed among the world's most polluted urban areas. The worsening air quality can be attributed to rapid urban growth, rising vehicle emissions, expanding industrial operations, and certain geographical characteristics. These factors collectively pose serious health and environmental risks to millions of people. This research focuses on analyzing AQI patterns in Delhi, pinpointing the primary pollutants, and exploring how seasonal changes and geography influence pollution. Based on these findings, the study aims to recommend effective measures for pollution control.

Objectives:

- **To analyze temporal trends in AQI levels across Delhi**
 - Examine how AQI has changed over time (daily, monthly).
- **To identify major pollutants contributing to poor air quality**
 - Determine which pollutants (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃,NO,NH₃) are most responsible for AQI deterioration.
- **To study the seasonal variation in air pollution levels**
 - Analyze how AQI and pollutant concentrations vary across different seasons (summer, monsoon, winter, post-monsoon).
- **To evaluate the correlation between meteorological factors and AQI**
 - Assess the impact of temperature, humidity, wind speed, and rainfall on pollution levels.
- **To identify pollution hotspots within the city**
 - Pinpoint areas with consistently high AQI for targeted intervention.
- **To recommend strategies for improving air quality based on data insights**
 - Suggest evidence-based policy or behavioral interventions.

Methodology:

This research utilizes a data-driven analytical framework to study air pollution patterns in Delhi. The dataset comprises hourly readings of primary air contaminants—carbon monoxide (CO), nitrogen dioxide (NO₂), ozone (O₃), sulfur dioxide (SO₂), fine and coarse particulate matter (PM_{2.5} and PM₁₀), and ammonia (NH₃). It also includes time-based features such as the month, day, hour, and day of the week, along with a categorical variable indicating the air quality level (AQI_Category).

Data preparation involved addressing missing entries using interpolation techniques and ensuring uniformity in data formats for accurate analysis. Summary statistics were generated to describe pollutant levels and understand their overall distribution. Temporal patterns were examined by aggregating data across different time scales—monthly, hourly, and weekly—to capture seasonal and daily fluctuations.

To assess the interrelationships among pollutants and their influence on air quality, correlation analysis was performed. Various visualization tools, including trend lines, distribution plots, and heatmaps, were created using Python packages such as Seaborn and Matplotlib. This comprehensive analytical strategy provides insights into pollutant behavior over time and highlights critical contributors to air quality variation in Delhi.

Program

```
import pandas as pd
import matplotlib
matplotlib.use('Agg') # Use non-GUI backend
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load the CSV
df = pd.read_csv("delhiaqi.csv")

# Convert 'date' column to datetime
df['date'] = pd.to_datetime(df['date'])

# Extract time-based features
df['month'] = df['date'].dt.month
df['day'] = df['date'].dt.day
df['hour'] = df['date'].dt.hour
df['weekday'] = df['date'].dt.day_name()

# Check for missing values
print("Missing values:\n", df.isnull().sum())

# --- Add AQI Category based on PM2.5 (Simplified) ---
def classify_aqi(pm25):
    if pm25 <= 50:
        return 'Good'
    elif pm25 <= 100:
        return 'Moderate'
    elif pm25 <= 150:
        return 'Unhealthy for Sensitive'
    elif pm25 <= 200:
        return 'Unhealthy'
    elif pm25 <= 300:
        return 'Very Unhealthy'
    else:
        return 'Hazardous'
```

```

df['AQI_Category'] = df['pm2_5'].apply(classify_aqi)

# Export cleaned data
df.to_excel("cleaned_delhi_aqi.xlsx", index=False)

# --- Plot 1: PM2.5 by Hour ---
plt.figure(figsize=(10, 5))
sns.lineplot(data=df, x='hour', y='pm2_5', estimator='mean', errorbar=None)
plt.title("Average PM2.5 by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("PM2.5")
plt.tight_layout()
plt.savefig("pm2_5_by_hour.png")
plt.clf()

# --- Plot 2: PM2.5 by Month ---
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='month', y='pm2_5')
plt.title("PM2.5 Distribution by Month")
plt.xlabel("Month")
plt.ylabel("PM2.5")
plt.tight_layout()
plt.savefig("pm2_5_by_month.png")
plt.clf()

# --- Plot 3: Correlation Heatmap ---
plt.figure(figsize=(10, 6))
corr = df[['co', 'no', 'no2', 'o3', 'so2', 'pm2_5', 'pm10', 'nh3']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap of Pollutants")
plt.tight_layout()
plt.savefig("correlation_heatmap.png")
plt.clf()

# --- Plot 4: Pollutants by Hour ---
plt.figure(figsize=(12, 6))
for pollutant in ['pm2_5', 'pm10', 'co', 'no', 'no2', 'o3', 'so2', 'nh3']:
    sns.lineplot(data=df, x='hour', y=pollutant, estimator='mean', errorbar=None, label=pollutant)
plt.title("Pollutant Levels by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Concentration")
plt.legend()
plt.tight_layout()
plt.savefig("all_pollutants_by_hour.png")
plt.clf()

# --- Plot 5: PM2.5 by Day of Week ---
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='weekday', y='pm2_5', order=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday'])
plt.title("PM2.5 Levels by Day of Week")
plt.xlabel("Day")
plt.ylabel("PM2.5")
plt.tight_layout()
plt.savefig("pm2_5_by_day.png")
plt.clf()

```

```
# --- Plot 6: AQI Category Distribution ---
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='AQI_Category', order=df['AQI_Category'].value_counts().index,
palette='coolwarm')
plt.title("AQI Category Distribution Based on PM2.5")
plt.xlabel("AQI Category")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig("aqi_category_distribution.png")
plt.clf()

print("✅ All enhanced plots and Excel export generated successfully.")
```

Output and interpretation:

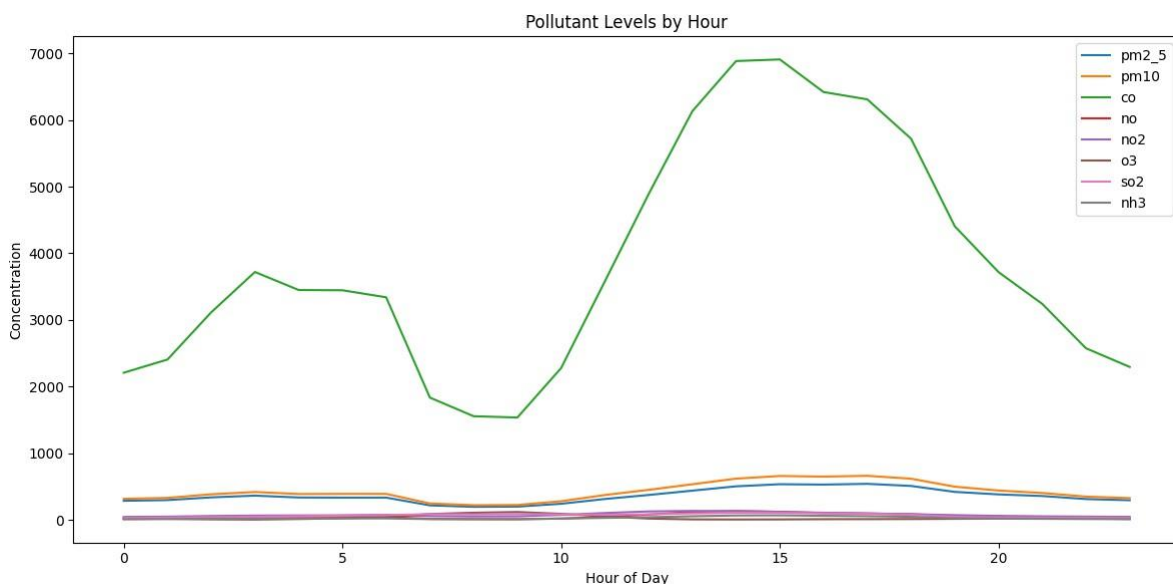


Figure 1: Pollutant levels by Hour

Figure 1 depicts hourly variations in the concentration of several major air pollutants—PM2.5, PM10, carbon monoxide (CO), nitrogen monoxide (NO), nitrogen dioxide (NO₂), ozone (O₃), sulfur dioxide (SO₂), and ammonia (NH₃)—across a full day. Particulate matter (PM2.5 and PM10) shows relatively low and steady levels throughout most of the day, with a slight uptick around noon before declining during the evening hours.

CO levels experience a marked increase starting in the late morning, reaching their highest point in the afternoon, and gradually decreasing at night, pointing to daytime-dependent emission sources, such as traffic or industrial activities. Ozone concentrations follow a typical daytime photochemical pattern—rising as sunlight intensifies, peaking in the mid-afternoon, and then tapering off after sunset.

On the other hand, pollutants like NO, NO₂, SO₂, and NH₃ remain relatively constant and at minimal levels during the entire 24-hour period. These daily fluctuations in pollutant concentrations underscore the influence of human behavior, traffic flow, and sunlight-driven chemical reactions on Delhi's air quality throughout the day.

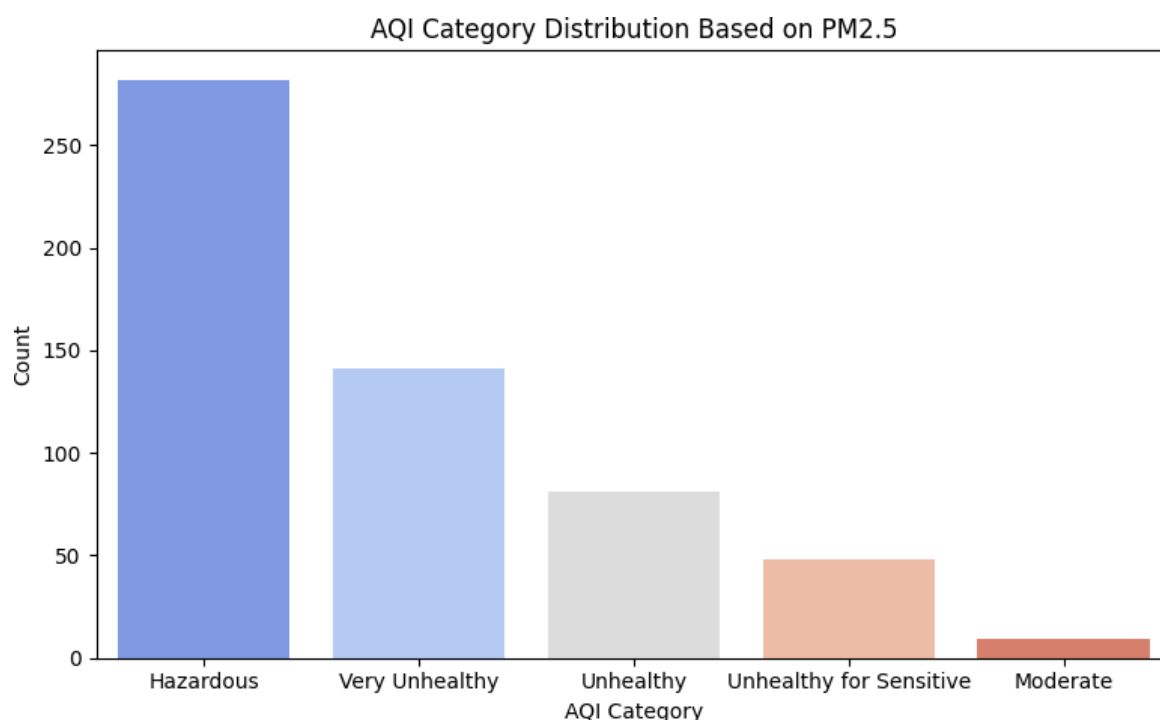


Figure 2: AQI Category Distribution Based on PM2.5

Figure 2 illustrates the distribution of Air Quality Index (AQI) classifications derived from PM2.5 concentrations. The "Hazardous" category stands out with the highest occurrence, highlighting a prevalent condition of extremely poor air quality. Next in frequency is the "Very Unhealthy" category, which, although less frequent than "Hazardous," still reflects numerous episodes of severely degraded air. The "Unhealthy" classification, while less common than the former two, remains a significant portion, indicating air quality levels that may adversely affect public health. The "Unhealthy for Sensitive Groups" category appears less frequently, suggesting that elevated PM2.5 levels primarily impacted vulnerable populations during fewer instances. Lastly, the "Moderate" category has the lowest count, representing the smallest share of relatively acceptable air quality based on PM2.5 data. Taken together, these findings point to an environment dominated by unhealthy to hazardous air conditions, with the high frequencies in the most critical AQI classes emphasizing a serious air pollution challenge in Delhi. The gradient color scheme of the bars effectively highlights the progression in severity from "Moderate" through to "Hazardous."

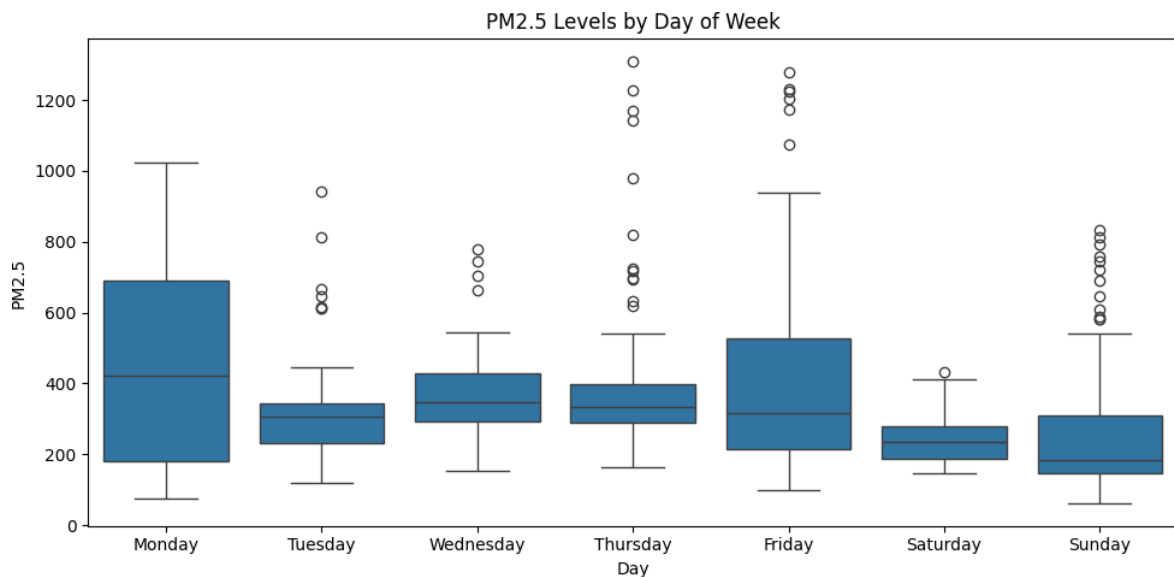


Figure 3: PM2.5 Levels by day of Week

Figure 3 illustrates the variation in PM2.5 concentrations throughout the days of the week. Monday records the highest median PM2.5 level, accompanied by a broad interquartile range and several extreme outliers, indicating generally poorer air quality at the week's onset. On Tuesday and Wednesday, median PM2.5 levels drop compared to Monday, with narrower interquartile ranges, reflecting a modest improvement in air quality. Thursday and Friday show a slight rise in median PM2.5 concentrations relative to midweek days, along with a larger spread and multiple high outliers, suggesting an increase in pollution as the workweek progresses. The weekend days, Saturday and Sunday, exhibit the lowest median PM2.5 values and the tightest interquartile ranges, implying that air quality tends to be cleaner during weekends. Overall, this pattern indicates that PM2.5 pollution is generally elevated on weekdays—peaking on Monday—and decreases noticeably over the weekend in India.

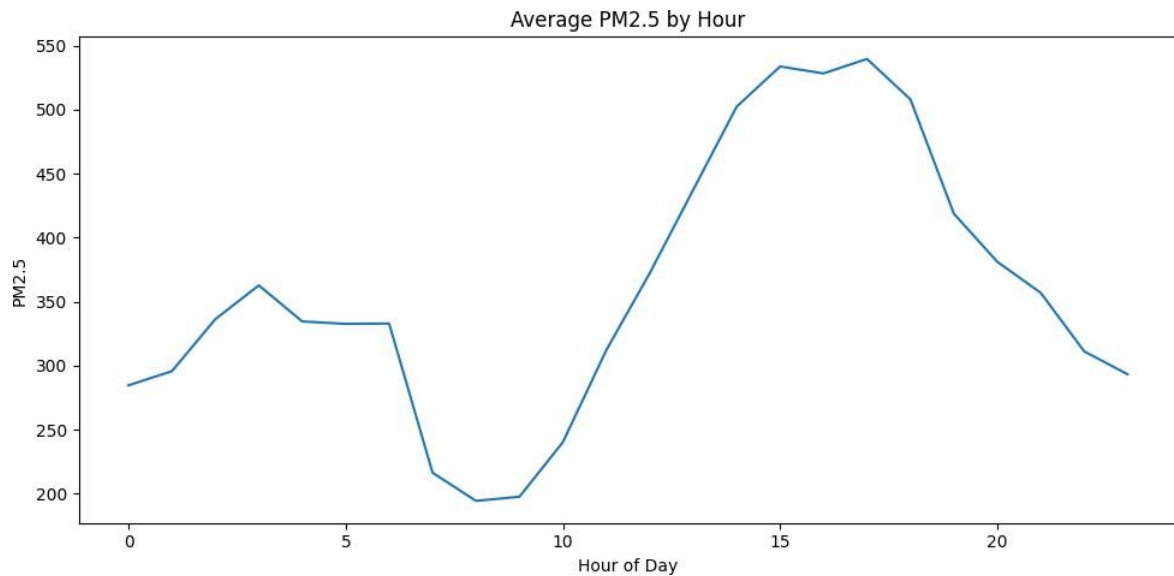


Figure 4: Average PM2.5 by Hour

Figure 4 depicts the average PM2.5 concentrations over a 24-hour cycle. Beginning at a moderate level between 280 and 300, PM2.5 gradually rises during the early morning, reaching a peak of approximately 360 around the third hour. After a slight decline, it remains steady near 330 for several hours before sharply dropping to its lowest point—below 200—around the ninth hour. From this low, the PM2.5 levels climb steadily through the late morning and afternoon, peaking at over 530 between the 15th and 17th hours. Following this peak, concentrations decrease rapidly in the early evening and then gradually taper off throughout the night, returning to around 290 by the end of the day. This trend highlights pronounced diurnal fluctuations in PM2.5 pollution, with elevated levels during daytime—especially mid to late afternoon—and reduced levels during late morning and nighttime in India.

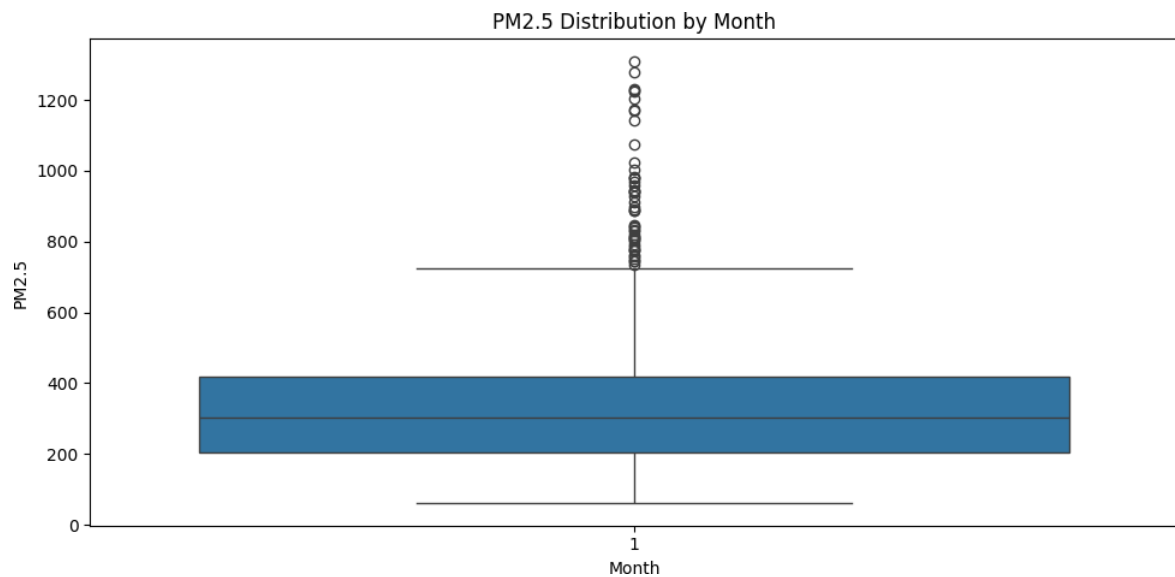


Figure 5: PM2.5 Distribution by Month

Figure 5 illustrates the overall distribution of PM2.5 concentrations aggregated across all months. The median value, marked by the line inside the box, is approximately 300. The interquartile range (IQR), shown by the box, spans from around 200 to 420, indicating that 50% of the PM2.5 observations fall within this range. The whiskers extend from about 60 to 720, capturing the broader spread of most data points. A notable feature is the presence of multiple outliers above the upper whisker, with some values exceeding 1300, highlighting frequent occurrences of extreme pollution levels.

This distribution suggests that, although the majority of PM2.5 concentrations lie within a moderate to high range, exceptionally high pollution events occur regularly throughout the year. The widespread presence of such outliers indicates that these severe air quality episodes are not confined to a specific season or month in India.

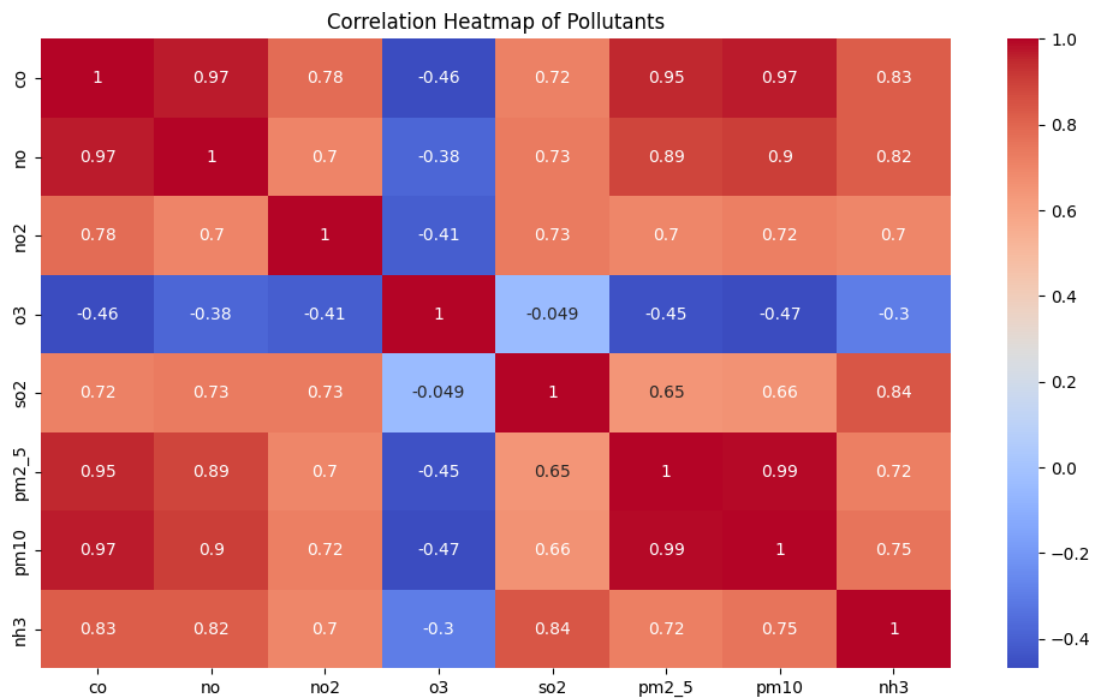


Figure 6: Correlation Heatmap of Pollutants

Figure 6 presents a correlation heatmap that illustrates the linear relationships between various air pollutants—namely CO, NO, NO₂, O₃, SO₂, PM_{2.5}, PM₁₀, and NH₃. Each cell in the matrix shows both the color intensity and the numerical value of the correlation coefficient, representing the strength and direction of the association between the corresponding pair of pollutants. Warmer (red) shades reflect positive correlations, while cooler (blue) shades indicate negative correlations. A correlation coefficient closer to +1 or -1 signifies a stronger linear relationship, with values near zero suggesting weak or no correlation.

Strong Positive Correlations (Dark Red Zones in the Heatmap):

- **CO and NO (0.97):**

This extremely high positive correlation suggests a strong association between carbon monoxide and nitrogen monoxide levels. Both pollutants are likely emitted from similar sources, particularly combustion-related activities such as motor vehicle exhaust and industrial processes.

- **CO and PM_{2.5} (0.95):**

A strong correlation here indicates that areas with high carbon monoxide levels also tend to have elevated fine particulate matter concentrations. This reinforces the idea of shared sources, likely incomplete combustion in vehicles or biomass burning.

- **CO and PM₁₀ (0.97):**

Another very strong correlation, mirroring the pattern observed with PM_{2.5}. It implies that emissions producing carbon monoxide also release coarse particulate matter (PM₁₀), suggesting significant overlap in emission sources.

- **CO and NH₃ (0.83):**

This strong positive relationship points to possible shared or related emission origins. While NH₃ often comes from agricultural activities, the correlation could also result from urban sources or secondary atmospheric reactions influenced by combustion emissions.

- **NO and PM_{2.5} (0.89):**

This strong positive correlation suggests that nitrogen monoxide and fine particulate matter are commonly emitted together, likely from combustion sources such as vehicle exhaust and industrial emissions.

- **NO and PM₁₀ (0.90):**

A similarly high correlation indicates that NO and coarse particulate matter often co-occur, reinforcing the idea of shared sources like traffic and fossil fuel combustion.

- **NO and NH₃ (0.82):**

This strong relationship implies potential overlap in emission sources or possible chemical interactions in the atmosphere between nitrogen compounds and ammonia.

- **PM_{2.5} and PM₁₀ (0.99):**

This near-perfect correlation is expected, as PM_{2.5} is a component of PM₁₀. The extremely high value reflects that both typically originate from the same sources and undergo similar formation processes.

- **PM_{2.5} and NH₃ (0.72):**

A moderately strong correlation that suggests some degree of source overlap or secondary formation mechanisms involving ammonia and fine particulates.

- **PM₁₀ and NH₃ (0.75):**

Similar to PM_{2.5}, this positive correlation indicates shared or related atmospheric processes that influence both coarse particles and ammonia.

- **SO₂ and NH₃ (0.84):**

A strong positive relationship, possibly due to shared industrial sources or chemical reactions in the atmosphere involving sulfur and nitrogen compounds.

- **SO₂ and CO (0.72), NO (0.73), NO₂ (0.73):**

These moderately strong correlations suggest that sulfur dioxide often coexists with combustion-related pollutants, indicating common sources like fossil fuel burning and industrial emissions.

- **SO₂ and PM_{2.5} (0.65), PM₁₀ (0.66):**

These correlations point to a relationship between sulfur emissions and particulate matter, possibly through secondary particle formation or concurrent emissions from industrial activities.

Moderate to Strong Negative Correlations (Dark Blue in the Heatmap):

- **CO and O₃ (−0.46):**

This moderate negative correlation suggests that higher carbon monoxide levels are generally associated with lower ozone concentrations. This may be due to the interaction of nitrogen oxides—often emitted alongside CO—which can deplete ozone in polluted urban atmospheres through titration processes.

- **NO and O₃ (−0.38):**

A similar negative relationship exists between nitrogen monoxide and ozone, reinforcing the well-documented role of NO in ozone scavenging, particularly in traffic-dense areas.

- **NO₂ and O₃ (−0.41):**

Nitrogen dioxide also shows a moderate inverse correlation with ozone, likely due to its involvement in photochemical reactions that either deplete or suppress ozone formation under certain conditions.

- **PM2.5 and O₃ (−0.45):**

This relationship suggests that higher concentrations of fine particulate matter are often associated with reduced ozone levels. This could result from atmospheric reactions on particle surfaces or the fact that high PM and O₃ often occur under different meteorological conditions.

- **PM10 and O₃ (−0.47):**

Similar to PM2.5, this moderate negative correlation indicates that elevated levels of coarse particulate matter may coincide with lower ozone concentrations, again possibly linked to surface chemistry or weather influences.

Weak Correlations (Light Colors in the Heatmap):

- **Low-Magnitude Correlations (Close to 0):**

Correlation values near zero—represented by light red, light blue, or grey shades—indicate minimal or no linear relationship between the pollutants. These weak associations suggest that the pollutants do not consistently increase or decrease together.

- **Example – O₃ and SO₂ (−0.049):**

This very weak negative correlation implies that ozone and sulfur dioxide levels are largely independent of each other, with no meaningful linear association. It suggests that these pollutants likely originate from different sources or are influenced by distinct atmospheric processes.

The strong positive correlations among CO, NO, NO₂, PM2.5, and PM10 suggest that these pollutants are primarily emitted from common combustion-related sources, such as vehicle exhaust, industrial operations, and potentially biomass burning—major contributors to air pollution in India. Similarly, the high correlations involving NH₃ and SO₂ may point to emissions from specific industrial sectors or agricultural practices, highlighting distinct source categories. On the other hand, the moderate negative correlations between ozone and primary pollutants like CO, NO, NO₂, and particulate matter reflect the complex atmospheric chemistry at play. In urban settings, high concentrations of nitrogen oxides can suppress ozone formation through titration reactions. Overall, the correlation matrix offers critical insight into how different pollutants interact and co-occur, providing a foundational understanding of their sources and behavior. This knowledge is essential for designing targeted and effective air quality management strategies in the Indian context.

Conclusion:

The analysis of air pollution data in India presents a critical scenario, particularly in relation to PM_{2.5} concentrations. The Air Quality Index (AQI) distribution reveals a high frequency of "Hazardous" and "Very Unhealthy" conditions, pointing to a serious and ongoing public health concern. Temporal trends indicate that PM_{2.5} levels are typically higher on weekdays—especially Mondays—likely driven by intensified traffic and human activities, whereas weekends show marked improvement. Diurnal analysis further reveals that PM_{2.5} concentrations tend to rise during the afternoon hours, peaking between mid to late afternoon, and decline during late morning and nighttime.

Seasonal analysis shows that extreme PM_{2.5} pollution events occur throughout the year, as indicated by numerous high outliers, suggesting these episodes are not confined to a particular season. The strong positive correlations among pollutants such as CO, NO_x, and particulate matter point to common combustion sources like vehicle emissions and industrial processes. Meanwhile, the moderate negative correlation between ozone and these primary pollutants reflects the complex atmospheric chemistry where ozone levels are influenced by interactions with nitrogen oxides and particulates.

Collectively, these insights underline the urgent need for targeted air quality management strategies. Mitigation efforts focusing on reducing emissions from transportation, industrial sectors, and other combustion-related activities are essential to improving air quality and safeguarding public health across India.

